

Scenario: Campus Printing & Copy Center App

Module Name: Programming 2

Module Code: CMP1025

Date: February 2026

Duration: 1½Hours

Lab Test 1: Python

Scenario: University Print Shop

Write a program that simulates a university print/copy center. A student can add print jobs to their order, choose the type of paper, and checkout. The program must use a menu that repeats until the user checks out successfully or chooses to exit.

The application keeps track of the student's current print order and calculates costs at checkout, including service charge and pickup fees. There is also a chance the student receives their order for free (promo).

Program Requirements

Main Menu

The program must repeatedly display the following menu until the user checks out successfully or chooses to exit:

1. Add print job
2. Select Paper
3. Checkout
4. Exit

Print Job Menu

When adding a print job, the user must be shown the following options:

- Black & White (per page): **\$15**
- Colour (per page): **\$40**
- Poster Print: **\$600**
- Binding: **\$300**

Note: When selecting Black & White or Colour, the user must enter the number of pages.

Select Paper

When selecting the paper type, show the following options:

- Plain: **\$20**
- Cartridge: **\$50**
- Photo: **\$500**

Checkout Rules

- Service charge: **10%** of the subtotal
- Base handling fee: **\$150**
- Pickup fee: **Added to the base handling fee**

Upon successful checkout:

- There is a **1 in 25** chance the entire order is free.
- If the order is free, the final total is **\$0.00**.

Required Functions

You must implement the following functions. Each function should perform only its intended task.

1. `add_print_job()`

- a. Displays the print job menu
- b. Allows the user to select a job
- c. If applicable, accepts quantity (pages)
- d. Returns the cost of the selected job

2. `select_paper_type()`

- a. Displays the paper options
- b. Allows the user to select the paper type
- c. Returns the cost of the selected item

3. `checkout(subtotal, pickup_fee)`

- a. Checks if the subtotal meets minimum requirements
- b. Calculates service charge, handling fee, and pickup fee
- c. Checks if the user gets the free promo
- d. Displays the final total when appropriate

Rubric (100 Marks)

Functions

- Correct Functionality
- Correct use of parameters and return values
- Accurate calculations

Function	Marks
add_print_job	15
Select_paper_type	15
checkout	20

Logic

- Correct use of loops and conditions
 - Accurate application of fees
 - Proper program flow and state management
- 40 marks**

Comments

- Name and ID clearly documented
 - Meaningful comments for major sections
- 10 marks**

Total: 100